Database Technologies – Assignment 9

Create appropriate indexes to solve the following problems.

Note: use “sales” database to solve below queries.

1. Create an index that will enable a user to pull orders for ‘1990-10-03’ out of the Orders table quickly.

```
KD3_chandrakant_96233>CREATE INDEX ord_odate_idx ON orders(odate);
Query OK, 0 rows affected (0.24 sec)
Records: 0  Duplicates: 0  Warnings: 0

KD3_chandrakant_96233>SELECT *
    -> FROM orders
    -> WHERE odate = '1990-10-03';
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+------+---------+------------+------+------+
5 rows in set (0.02 sec)

KD3_chandrakant_96233>
```

2. If the Orders table has already been created, how can you force the onum field to be unique (assume all current values are unique)?

```
KD3_chandrakant_96233>CREATE UNIQUE INDEX idx_orders_onum ON Orders(onum);
Query OK, 0 rows affected (0.08 sec)
Records: 0  Duplicates: 0  Warnings: 0

KD3_chandrakant_96233>DESCRIBE orders;
+-------+--------------+------+-----+---------+-------+
| Field | Type         | Null | Key | Default | Extra |
+-------+--------------+------+-----+---------+-------+
| onum  | int          | YES  | UNI | NULL    |       |
| amt   | decimal(7,2) | YES  |     | NULL    |       |
| odate | date         | YES  | MUL | NULL    |       |
| cnum  | int          | YES  |     | NULL    |       |
| snum  | int          | YES  |     | NULL    |       |
+-------+--------------+------+-----+---------+-------+
5 rows in set (0.05 sec)

KD3_chandrakant_96233>
```

3. Create an index that would permit salesperson to retrieve his orders.

```
KD3_chandrakant_96233>CREATE INDEX idx_orders_snum ON orders(snum);
Query OK, 0 rows affected (0.08 sec)
Records: 0  Duplicates: 0  Warnings: 0

KD3_chandrakant_96233>SELECT *
    -> FROM orders o
    -> WHERE o.snum IN (SELECT s.snum FROM salespeople s);
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3007 |   75.75 | 1990-10-04 | 2004 | 1002 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
+------+---------+------------+------+------+
10 rows in set (0.04 sec)
```

4. Let us assume that each salesperson is to have only one customer of a given rating, and this is currently the case. Create an index that enforces it.

```
KD3_chandrakant_96233>CREATE UNIQUE INDEX snum_rating_csum_idx
    -> ON customers(snum, rating);
ERROR 1062 (23000): Duplicate entry '1001-100' for key 'customers.snum_rating_csum_idx'
KD3_chandrakant_96233>
```

5. Create an index to speed up searching order on a given date by given customer.

```
ERROR 1062 (23000): Duplicate entry '1001-100' for key 'customers.snum_rating_csum_idx'
KD3_chandrakant_96233>CREATE UNIQUE INDEX _check_kar_idx
    -> ON orders(odate, cnum);
ERROR 1062 (23000): Duplicate entry '1990-10-03-2008' for key 'orders._check_kar_idx'
KD3_chandrakant_96233>
```